

Extra Credit Exercises for Module 1

- **Submit a complete solution and analysis, detailing all steps clearly.**
- **All calculations must be done by hand.**
- Please upload a single, high-quality PDF file containing clear pictures of your handwritten work.
- Each exercise is worth 2.5 points.

Exercise 1

- Perform two steps of gradient descent starting from the origin ($\mathbf{b}^{(0)} = (0, 0), c^{(0)} = 0$) for a linear regression problem in $\mathbb{R}^2$ with the following four data points:

$$\begin{aligned}\mathbf{x}_1 &= (1, 1), & y_1 &= 1, \\ \mathbf{x}_2 &= (1, 2), & y_2 &= 2, \\ \mathbf{x}_3 &= (2, 1), & y_3 &= 2, \\ \mathbf{x}_4 &= (2, 2), & y_4 &= 3.\end{aligned}$$

The goal is to minimize the mean squared error (MSE) for the linear model:

$$f(\mathbf{x}) = \mathbf{b}^T \mathbf{x} + c = b_1 x_1 + b_2 x_2 + c,$$

where $\mathbf{b} = (b_1, b_2)$ is the parameter vector, and c is the bias. Suppose the learning rate is 0.1 in all the iterations!

- Perform two steps of mini-batch gradient descent with a batch size of 2:
 - First batch: Points $(\mathbf{x}_1, \mathbf{x}_3)$,
 - Second batch: Points $(\mathbf{x}_2, \mathbf{x}_4)$.
- Perform two steps of gradient descent using a single point at each iteration:
 - First iteration: Use the third point $\mathbf{x}_3$,
 - Second iteration: Use the second point $\mathbf{x}_2$.

For each scenario, compute and update the parameters $\mathbf{b}$ and c over the two iterations.

Exercise 2

You are given the following linear programming problem:

Objective Function:

$$\text{Maximize } -x_1 + 2x_2$$

Constraints:

$$\begin{aligned} 2x_1 + x_2 &\leq 3 \\ x_1 &\geq 0, \quad x_2 \geq 0 \end{aligned}$$

Solve this linear program using Karush-Kuhn-Tucker (KKT) conditions. You do not need to find the dual problem!

Exercise 3

You are given the following linear programming problem:

Objective Function:

$$\text{Maximize } 3x_1 + 4x_2$$

Constraints:

$$\begin{aligned} x_1 + 2x_2 &\leq 5 \\ 3x_1 + x_2 &\leq 7 \end{aligned}$$

Find the dual of this linear programming problem. You do not need to solve the problem itself or its dual!

Exercise 4

You are given the following quadratic programming problem:

Objective Function:

$$\text{Minimize } \frac{1}{2} (4x_1^2 + x_2^2) + x_1 + x_2$$

Constraints:

$$\begin{aligned} x_1 + x_2 &\leq 2 \\ x_2 &\geq 1 \end{aligned}$$

Solve this quadratic program using Karush-Kuhn-Tucker (KKT) conditions. You do not need to find the dual problem!

Exercise 5

You are given the following quadratic programming problem:

Objective Function:

$$\text{Minimize } \frac{1}{2} (x_1^2 + 2x_2^2) - x_1 - 2x_2$$

Constraints:

$$x_1 + x_2 = 4$$

$$x_1 \geq 0$$

Find the dual problem. You do not need to solve the problem itself or its dual!

Exercise 6

You are given the following symmetric matrix:

$$A = \begin{pmatrix} 4 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 1 \end{pmatrix}$$

Compute the eigenvalues to determine its positive definiteness!